IWG 222 – FUNDAMENTOS DE INGENIERÍA DE CONSTRUCCIÓN Y ADMINISTRACIÓN

Profesor Responsable

Dr. Leonhard E. Bernold



Ayudantes

Catalina Agloni

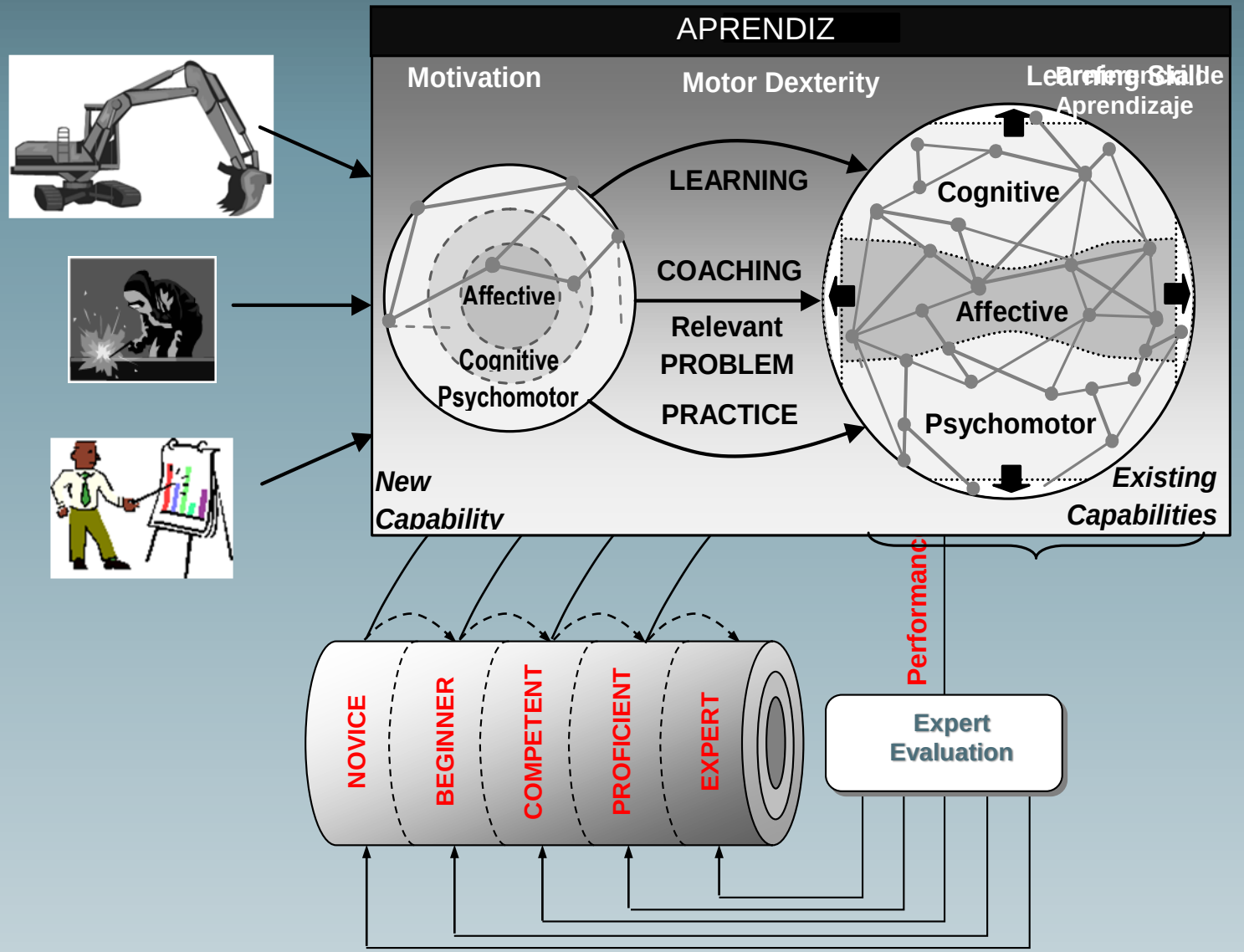
Fernando Leyton



CONTENIDO

- Capacitación del aprendiz
- Tecnologías para salva vidas
- Grúas - Los gigantes silenciosos

Ciclos para ganar competencia

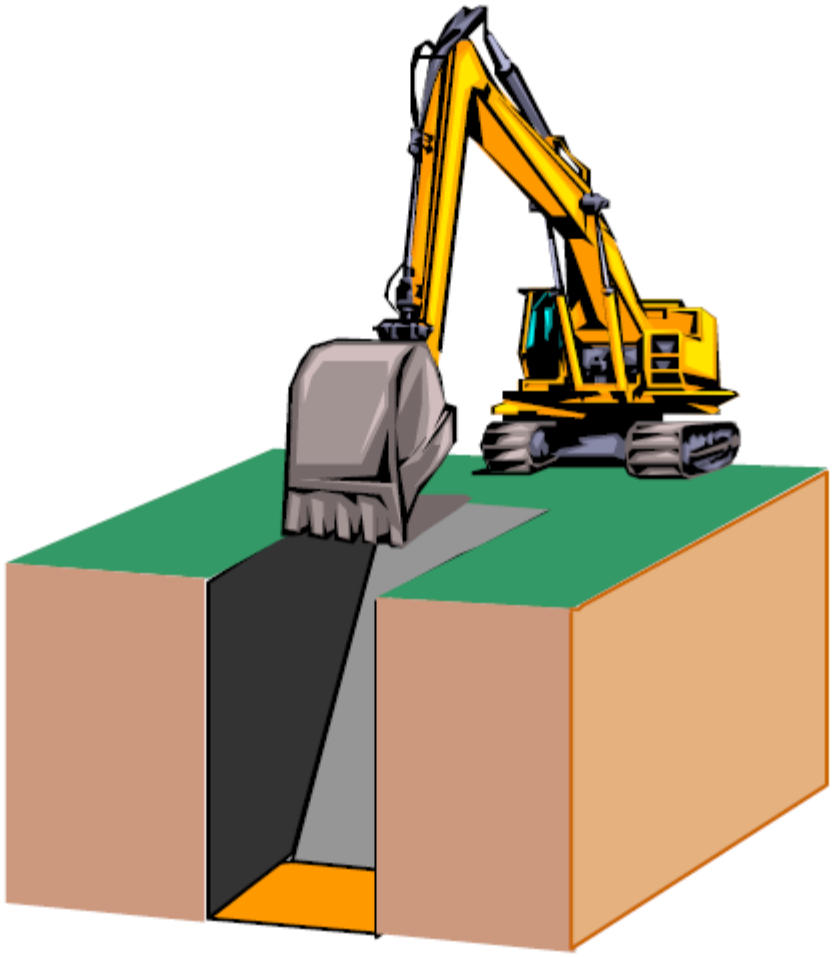



FUNDACION
LABORAL
DE LA CONSTRUCCION
0:19 / 3:28

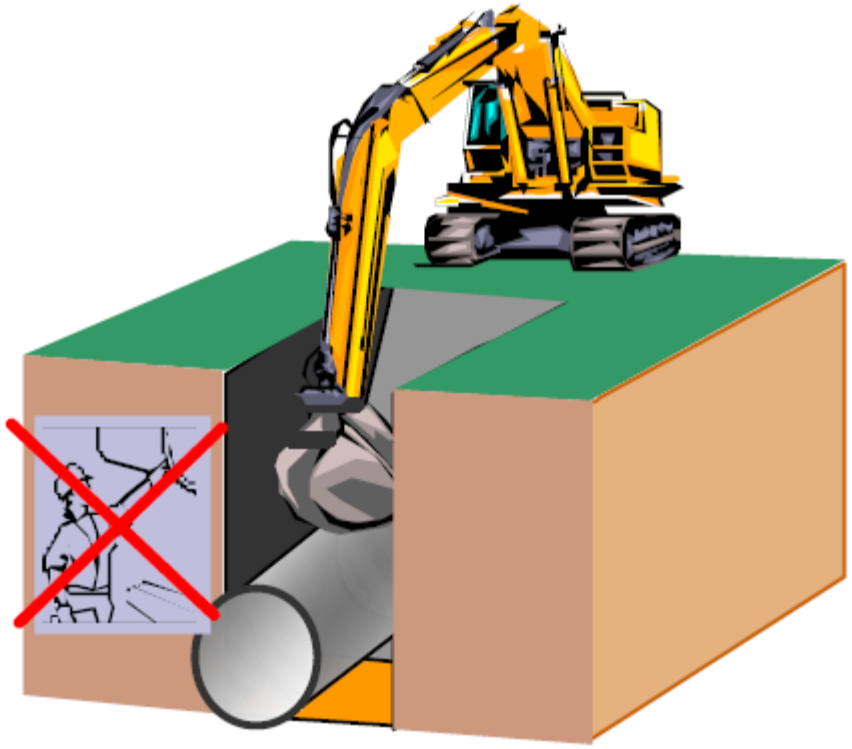








a) Trenching



b) Remote Pipe-Installation



a) First Prototype PipeMan



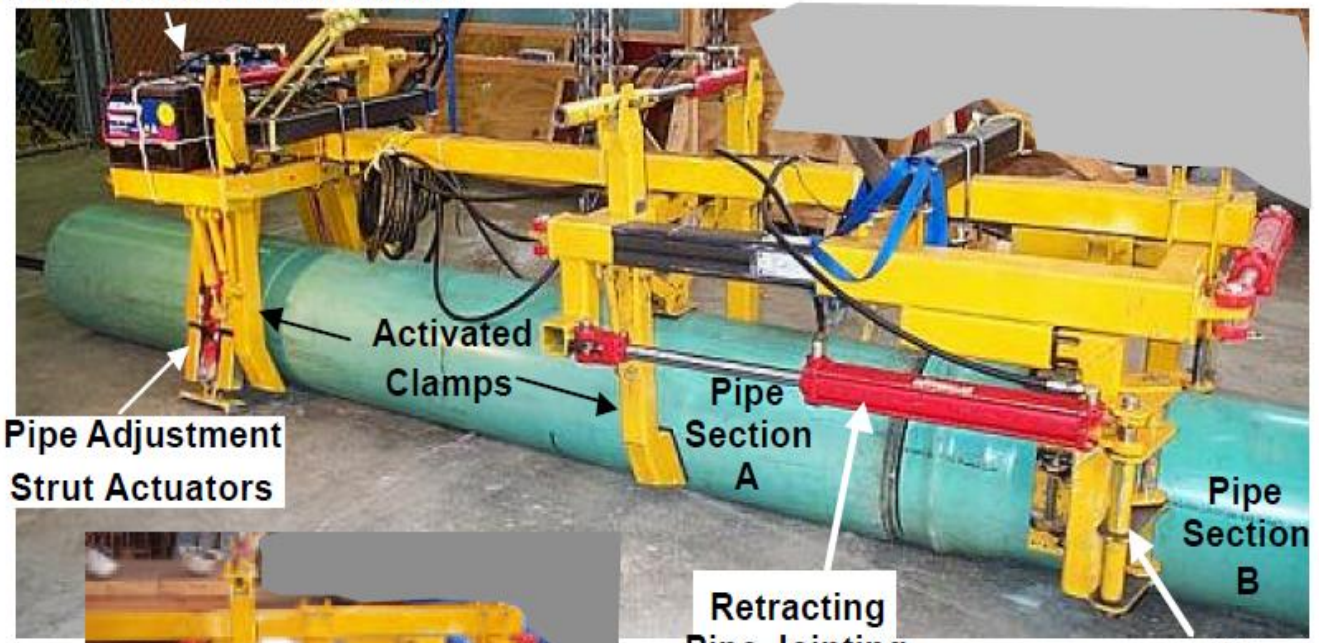
b) Second Generation in Field Test



c) Third Generation

Wireless Controls and Valves







Jointing in Progress

Grúas - Los gigantes silenciosos



























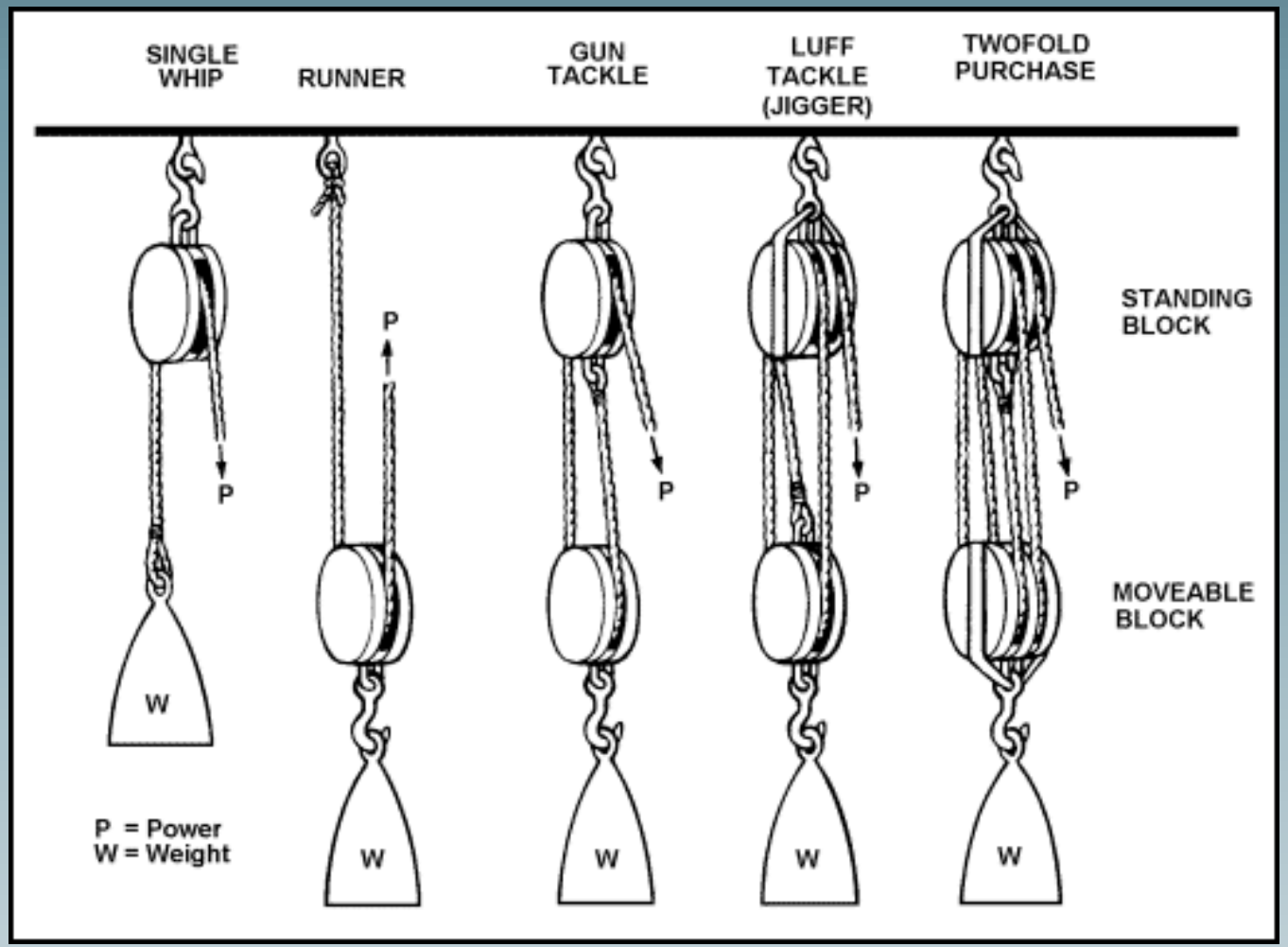


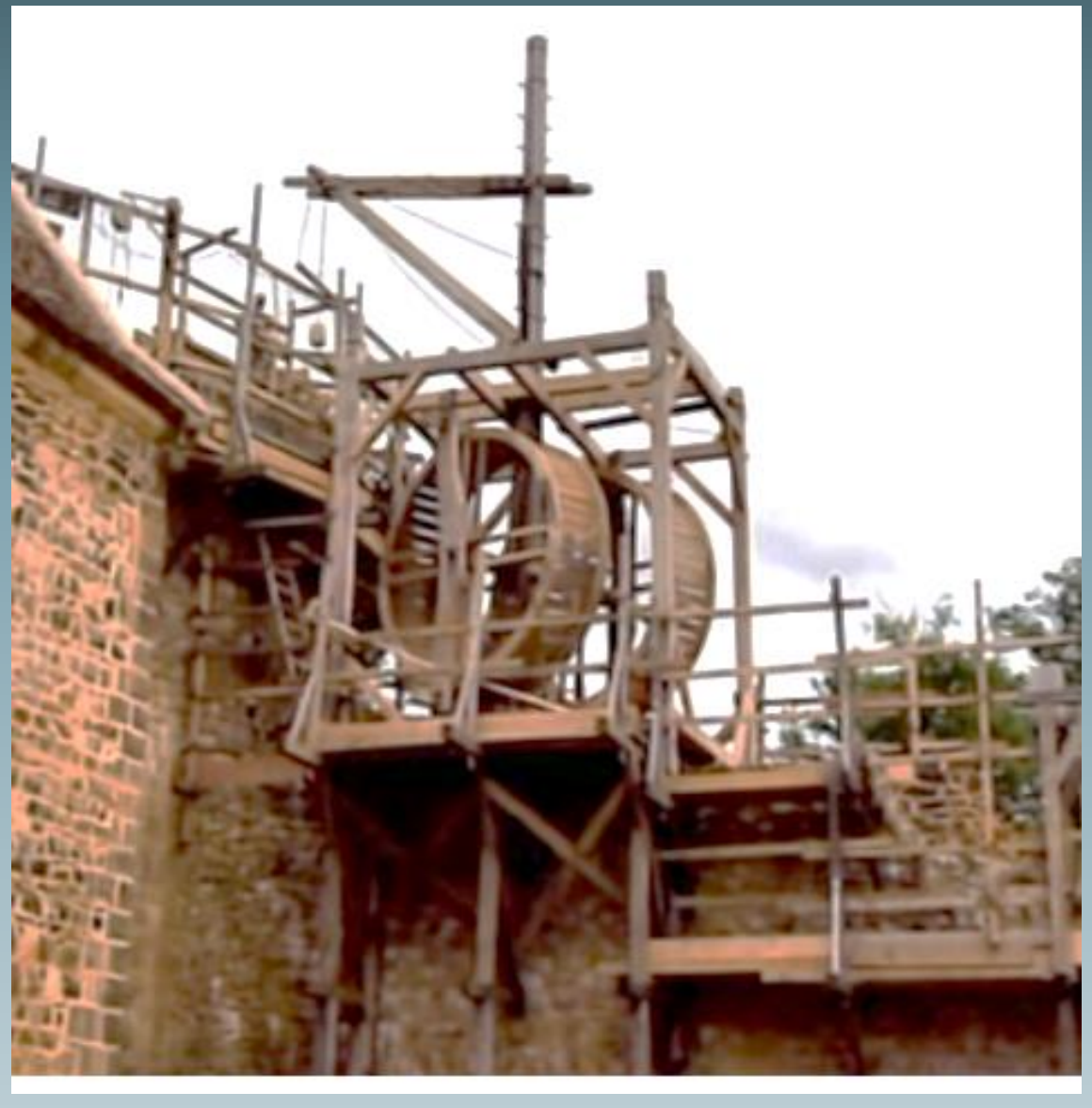
Camión Grúas





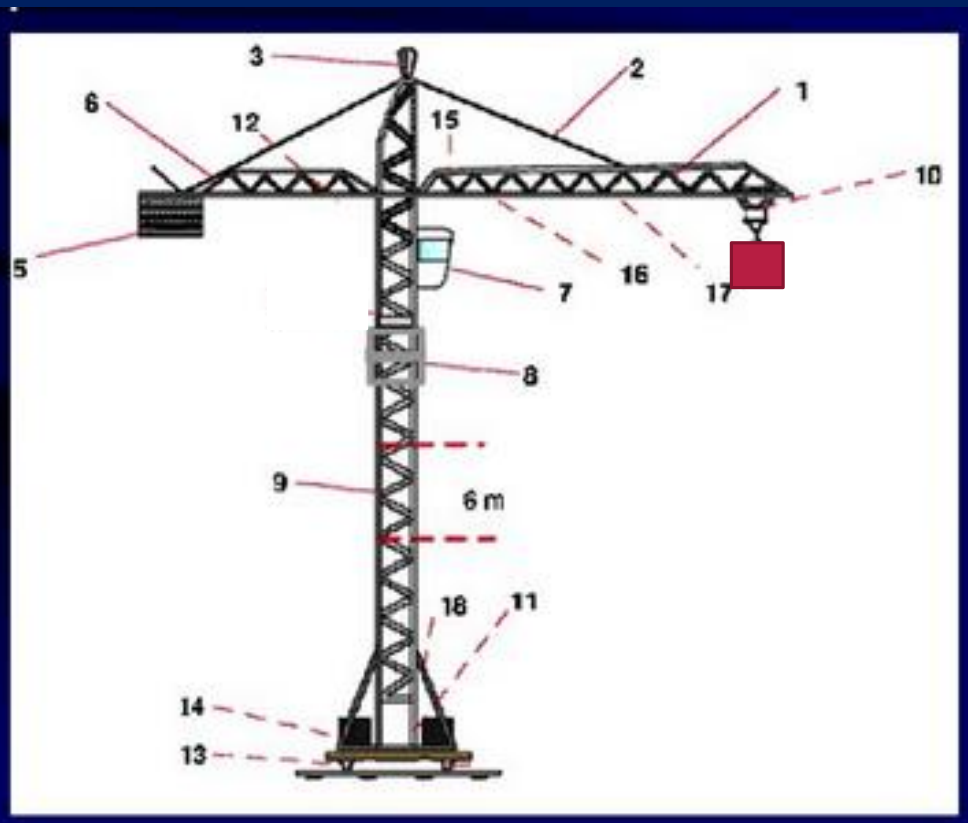
Archimedes


SINGLE WHIP
RUNNER
GUN TACKLE
LUFF TACKLE (JIGGER)
TWOFOLD PURCHASE
STANDING BLOCK
MOVEABLE BLOCK
P
W
P = Power
W = Weight



https://youtu.be/6zaHfL8rI5Y

Grúas Torre

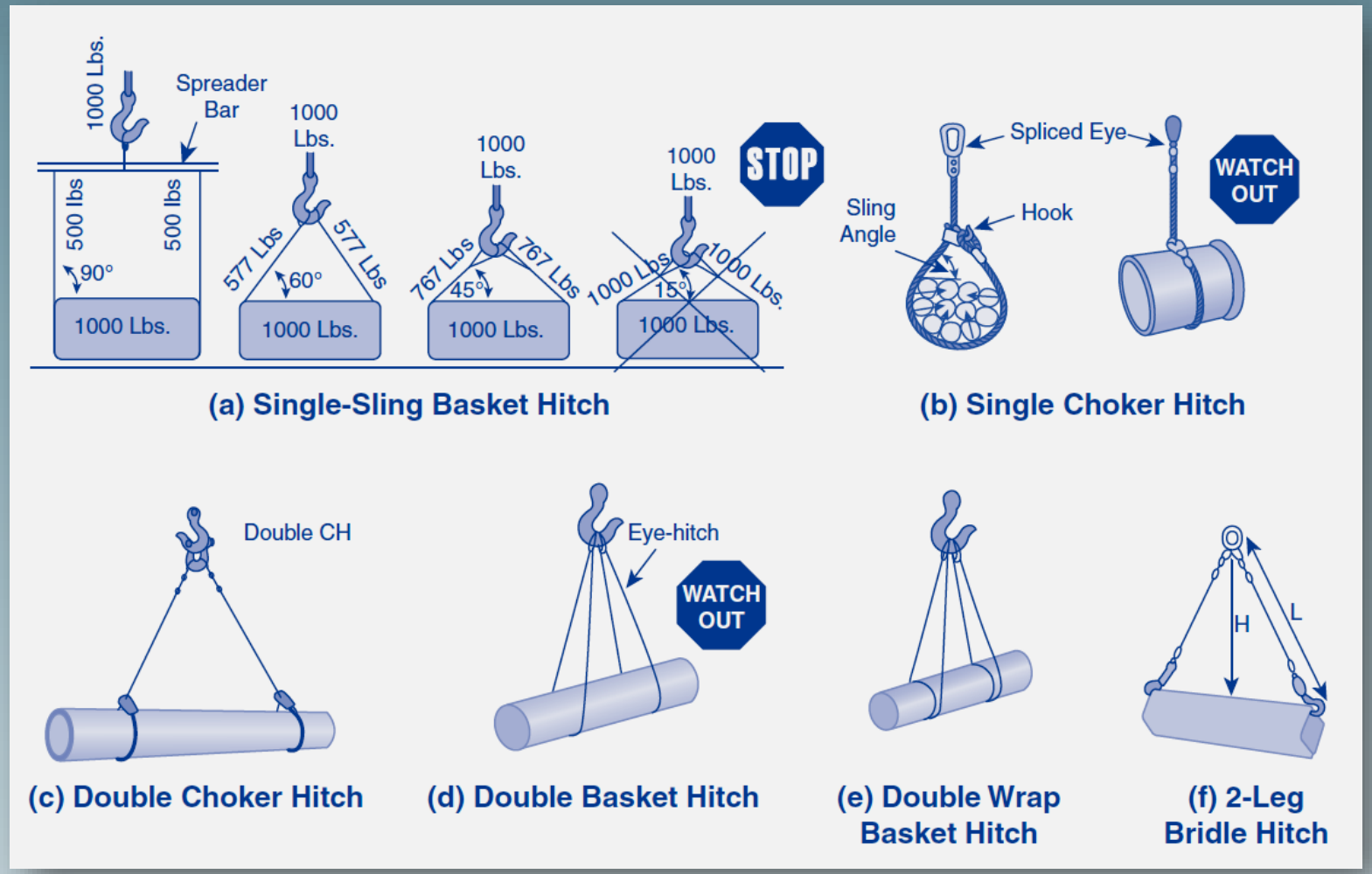


1.-Pluma
2.-Tensores
3.-Cabeza de Torre
4.-Corona de Giro
5.-Contrapeso Superior
6.-Contrapluma
7.-Cabina de Mando
8.-Tramo Deslizante
9.-Troncos Intermedios
10.-Gancho y Carro
11.-Tramo basal
12.-Motor Elevación
13.-Chasis
14.-Lastre Basal
15.-Motor de Giro
16.-Motor de Carro
17.-Cable de Carro
18.-Diagonales



https://youtu.be/HxjJssbqnJo

Equipos de aparejo – Enganche con la eslinga



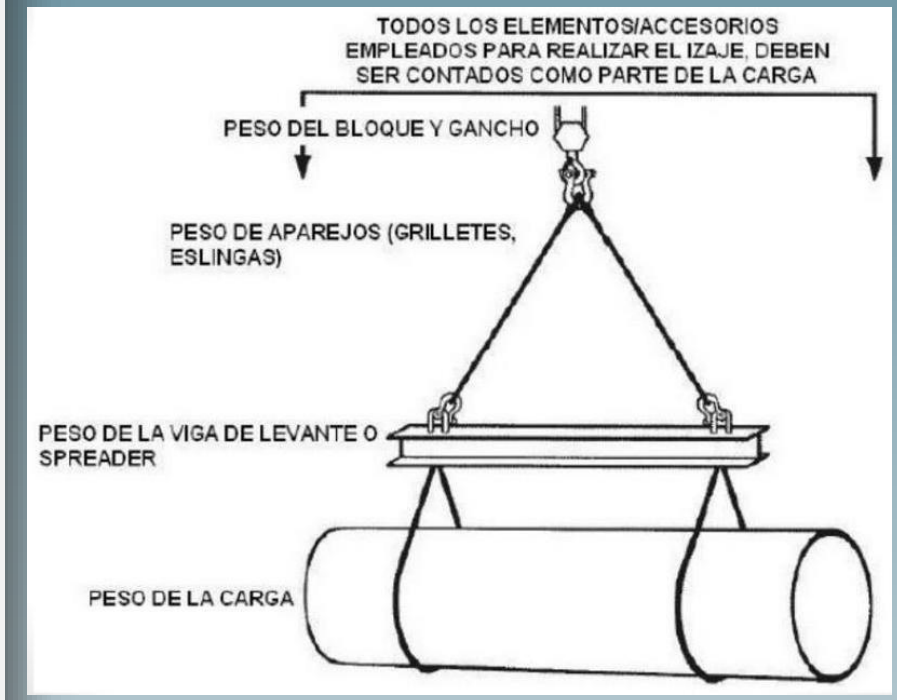
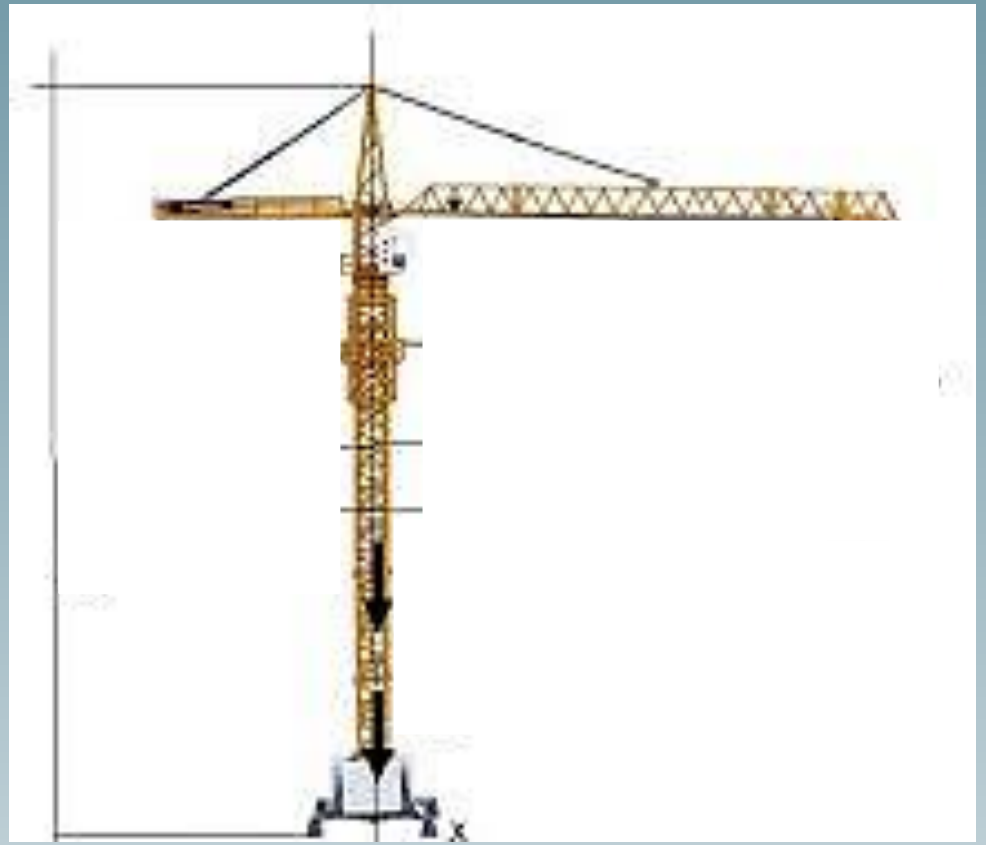
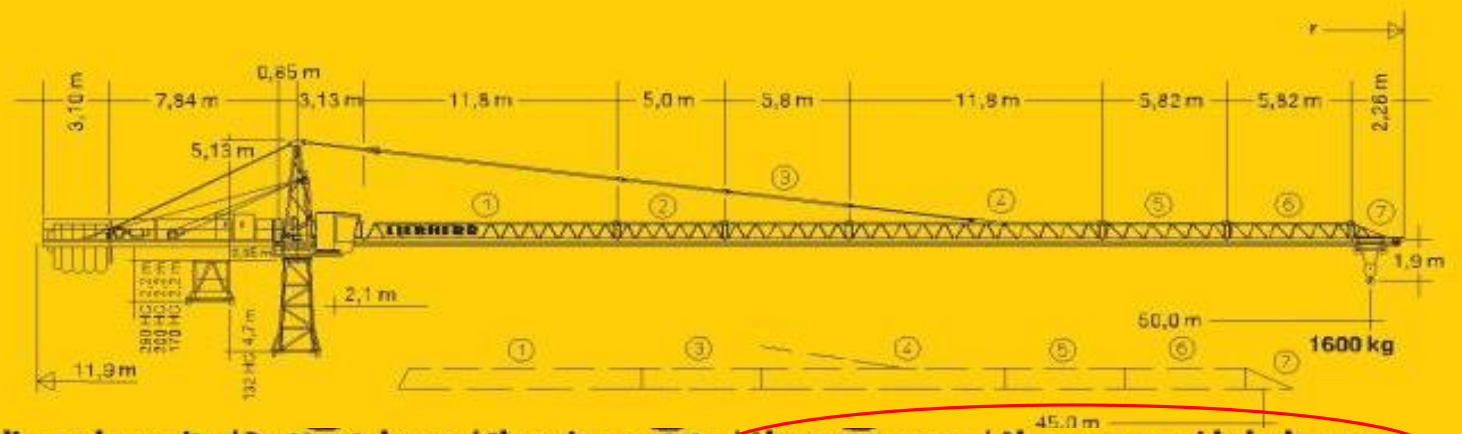


Diagrama de cuerpo libre

¿Efecto de la carga en movimiento?





Radius and capacity / Portée et charge / Sbraccio e portata / Alcances y cargas / Alcance e capacidade de carga

m r	m/kg	m/kg 18,0	20,0	22,0	24,0	27,6	30,0	33,4	36,0	39,2	42,0	45,0	48,0	50,0
50,0 (r = 51,5)	2,4 – 17,4 6000	5750	5070	4530	4080	3440	3110	2720	2470	2220	2030	1850	1690	1600
45,0 (r = 46,5)	2,4 – 18,7 6000	6000	5550	4960	4470	3780	3420	3000	2730	2450	2240	2050		
39,2 (r = 40,7)	2,4 – 19,6 6000	6000	5860	5240	4720	4000	3620	3170	2890	2600				
33,4 (r = 34,9)	2,4 – 20,2 6000	6000	6000	5440	4910	4150	3760	3300						
27,6 (r = 29,1)	2,4 – 20,8 6000	6000	6000	5620	5070	4300								

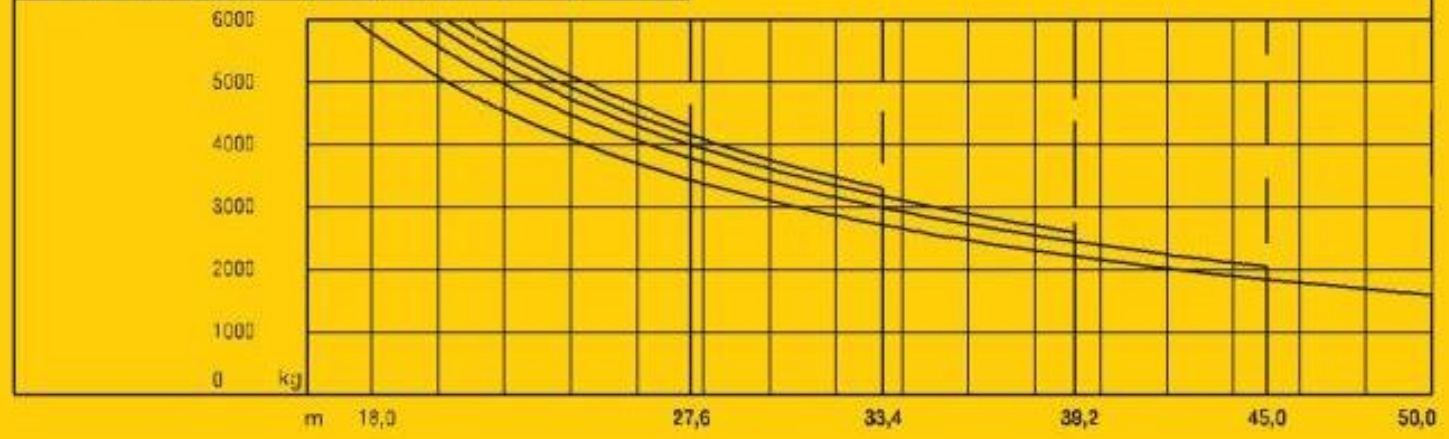


Table 7.4 Maximum Boom Load Capacities in Lb (t) for a Mobile Crane with 8,500 Lb (3.8 t) of Counterweight and Fully Extended Outriggers

Radius (ft)	Boom Length (ft)										
	38	50	60	70	76.5	90	100	110	120	130	140
10	154,600	152,100	138,900	81,700							
20	65,100	63,500	62,200	60,900	58,500	50,600	46,000	40,800			
30	41,900	41,700	41,600	40,600	40,100	39,600	38,200	37,300	33,800	29,800	22,500
40		29,100	28,400	27,900	28,100	28,000	28,000	27,500	27,100	26,900	20,000
50			21,100	20,600	20,400	20,200	20,100	19,900	19,600	19,100	16,100
60				15,400	15,000	14,600	14,500	14,300	14,000	13,600	12,700
70					10,800	11,000	10,900	10,800	10,400	10,100	9,700
80						8,400	8,300	8,200	7,900	7,500	7,200
90							6,300	6,300	6,000	5,600	5,300
100								4,800	4,500	4,100	3,800
110									3,300	3,000	2,700
120										2,000	1,700
130											900

Radius (m)	Boom Length (m)										
	11.58	15.24	18.29	21.34	23.32	27.43	30.48	33.53	36.58	39.62	42.67
3.0	70.13	68.99	63.00	37.06							
6.1	29.53	28.80	28.21	27.62	26.54	22.95	20.87	18.51			
9.1	19.01	18.91	18.87	18.42	18.19	17.96	17.33	16.92	15.33	13.52	10.21
12.2		13.20	12.88	12.66	12.75	12.70	12.70	12.47	12.29	12.20	9.07
15.2			9.57	9.34	9.25	9.16	9.12	9.03	8.89	8.66	7.30
18.3				6.99	6.80	6.62	6.58	6.49	6.35	6.17	5.76
21.3					4.90	4.99	4.94	4.90	4.72	4.58	4.40
24.4						3.81	3.76	3.72	3.58	3.40	3.27
27.4							2.86	2.86	2.72	2.54	2.40
30.5								2.18	2.04	1.86	1.72
33.5									1.50	1.36	1.22
36.6										0.91	0.77
39.6											0.41

= Load Limits Due to Possible Structural Failure
= Load Limits Due to Crane Tipping

Carga máximas basada de límites estructural o riesgo de volcar



JUNTA DE ANDALUCIA

CONSEJERÍA DE ECONOMÍA, INNOVACIÓN, CIENCIA Y EMPLEO
Dirección General de Industria, Energía y Minas

Firma

Apellidos: _____

Nombre: _____ **DNI/NIE** _____

15	A	**B**	C	D
	A	B	C	D

16	A	B	C	**D**
	A	B	C	D
17	A	**B**	C	D
	A	B	C	D
18	A	B	C	**D**
	A	B	C	D
19	**A**	B	C	D
	A	B	C	D
20	**A**	B	C	D
	A	B	C	D
21	A	**B**	C	D
	A	B	C	D
22	A	**B**	C	D
	A	B	C	D
23	A	B	**C**	D
	A	B	C	D
24	A	**B**	C	D
	A	B	C	D
25	A	**B**	C	D
	A	B	C	D
26	A	B	C	**D**
	A	B	C	D
27	**A**	B	C	D
	A	B	C	D
28	A	B	C	**D**
	A	B	C	D
29	**A**	B	C	D
	A	B	C	D
30	A	B	**C**	D
	A	B	C	D